

Final Project Submission and Presentation

Project Title

CoreTech Business Performance Dashboard

Problem Statement

CoreTech Innovation manages multiple clients, services, projects, and payments. Without proper data analysis, it becomes difficult to monitor business performance, identify high-performing services, track project progress, and make informed business decisions. This project aims to analyze business data and present meaningful insights through an interactive dashboard.

Project Objective

The objective of this project is to analyze CoreTech Innovation's business data using Microsoft Excel, MySQL, and Power BI to create an interactive dashboard that supports business decision-making and performance monitoring.

Dataset Description

The dataset contains 50 client records with the following fields:

- Client ID
- Client Name
- Company Name
- Service
- Project Budget
- Project Status
- Start Date
- End Date
- Client City
- Lead Source
- Payment Status

- Client Rating

The dataset was created to represent real business operations and client project information.

Dataset

A	B	C	D	E	F	G	H	I	J	K	L
Client_ID	Client_Name	Company_Name	Service	Project_Budget	Project_Status	Start_Date	End_Date	Client_City	Lead_Source	Payment_Status	Client_Rating
C001	Parkash	FBR	Web Development	5000	Completed	1/1/2026	1/3/2026	Hyderabad	Website	Paid	4.5
C002	Sagar	NovaTech Solutions	Data Analytics	7000	In Progress	3/1/2026	5/4/2026	Karachi	LinkedIn	Paid	4.2
C003	Ajina	Apex Innovations	Mobile App Development	6000	Pending	5/1/2026	20/3/2026	Islamabad	Referral	Pending	4.8
C004	Hina	BrightEdge Technologies	SEO Services	2500	In Progress	7/1/2026	15/4/2026	Lahore	Facebook	Pending	4.1
C005	Ahsas	CoreTech	Graphic Design	3500	Completed	10/1/2026	20/2/2026	Hala	Instagram	Paid	4.6
C006	Shane Ali	DigitalSphere Ltd	UI/UX Design	8000	In Progress	13/1/2026	25/4/2026	Sukkur	Website	Pending	4.7
C007	Ruman	NovaTech Solutions	Graphic Design	5500	Pending	15/1/2026	1/4/2026	Nawabshah	LinkedIn	Paid	4.3
C008	Emaan	CoreTech	Web Development	4000	In Progress	17/1/2026	5/5/2026	Larkana	Referral	Pending	4
C009	Nade	BrightEdge Technologies	UI/UX Design	4400	Completed	20/1/2026	10/5/2026	Hyderabad	Facebook	Paid	4.9
C010	Salar	Apex Innovations	Data Analytics	9000	In Progress	23/1/2026	23/2/2026	Karachi	Instagram	Pending	3.6
C011	Neelam	BrightFuture Tech	Web Development	6500	Completed	25/1/2026	25/2/2026	Islamabad	Facebook	Pending	3.9
C012	Nabeel	BrightEdge Technologies	Data Analytics	8500	In Progress	1/2/2026	10/5/2026	Lahore	Website	Pending	4.1
C013	Isha	NovaTech Solutions	Mobile App Development	5000	Pending	3/2/2026	15/5/2026	Hala	LinkedIn	Paid	4.3
C014	Nidesh	Spark Digital	SEO Services	3000	In Progress	5/2/2026	20/4/2026	Sukkur	Referral	Pending	4.2
C015	Harsha	Apex Innovations	Graphic Design	9200	Completed	7/2/2026	25/5/2026	Nawabshah	Facebook	Paid	5.5
C016	Saad	CoreTech	UI/UX Design	3500	Pending	10/2/2026	18/4/2026	Karachi	Instagram	Pending	4.8
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C018	Laiba	Vertex Digital	Graphic Design	2500	In Progress	15/2/2026	3/5/2026	Nawabshah	LinkedIn	Pending	3.5
C019	Bilal	NovaTech Solutions	Web Development	9700	Pending	17/2/2026	10/5/2026	Hyderabad	Facebook	Paid	3.7
C020	Maham	BrightEdge Technologies	Web Development	3400	In Progress	20/2/2026	17/5/2026	Karachi	Instagram	Pending	3.8
C021	Iqra	DigitalSphere Ltd	Data Analytics	2700	Completed	23/2/2026	20/6/2026	Islamabad	Facebook	Paid	3.9

Tools Used

- Microsoft Excel – Data creation, formulas, data cleaning, pivot tables, and charts
- MySQL – SQL queries and data analysis
- Power BI Desktop – KPI dashboard and business dashboard development

Analysis Process

The project was completed in the following steps:

1. Created the CoreTech business dataset.
2. Applied Excel formulas for business calculations.
3. Cleaned the dataset by handling missing values, removing duplicates, formatting dates and currency, and standardizing text.
4. Created pivot tables to summarize business information.
5. Built charts for data visualization.
6. Performed SQL queries for business analysis.
7. Developed Power BI dashboards with KPIs and interactive visuals.
8. Generated business insights and recommendations.

Data Cleaning

The dataset was cleaned to improve data quality and accuracy. Missing values were identified and corrected, duplicate records were checked and removed, date columns were properly formatted, project budgets were converted into currency format, and text values were standardized to maintain consistency.

Missing Values Handling

	A	B	C	D	E	F	G	H	I	J	K	L
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3	C002		NovaTech Solutions	Data Analytics	7000	In Progress	3/1/2026	5/4/2026	Karachi	LinkedIn	Paid	4.2
4	C003	Ajna	Apex Innovations	Mobile App Development	6000	Pending	5/1/2026	20/3/2026	Islamabad	Referral	Pending	4.8
5	C004	Hina		SEO Services	2500	In Progress	7/1/2026	15/4/2026	Lahore	Facebook	Pending	4.1
6	C005	Ahsas	CoreTech	Graphic Design	3500	Completed	10/1/2026	20/2/2026	Hala	Instagram	Paid	4.6
7	C006	Shane Ali	DigitalSphere Ltd	UI/UX Design		In Progress	13/1/2026	25/4/2026	Sukkur	Website	Pending	4.7
8	C007	Ruman	NovaTech Solutions	Graphic Design	5500	Pending	15/1/2026	1/4/2026	Nawabshah	LinkedIn	Paid	4.3
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10	C009	Nade	BrightEdge Technologies	UI/UX Design	4400	Completed	20/1/2026	10/5/2026	Hyderabad	Facebook	Paid	
11	C010	Salar	Apex Innovations	Data Analytics	9000	In Progress	23/1/2026	23/2/2026	Karachi	Instagram	Pending	3.6
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15	C014	Nidesh	Spark Digital	SEO Services	3000	In Progress	5/2/2026	20/4/2026	Sukkur	Referral	Pending	4.2
16	C015	Harsha	Apex Innovations	Graphic Design	9200	Completed	7/2/2026	25/5/2026	Nawabshah	Facebook	Paid	5.5
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23	C022	Sheeraz	BrightFuture Tech	Mobile App Development	8800	In Progress	25/2/2026	23/5/2026	Lahore	Instagram	Pending	4
24	C023	Hitik	FBR	SEO Services	9500	Completed	1/3/2026	5/5/2026	Hala	Referral	Paid	4.9
25	C024	Disha	SmartEdge Technologies	Graphic Design	9000	Pending	3/3/2026	10/5/2026	Sukkur	Website	Pending	4.2
26	C025	Madhu	Spark Digital	UI/UX Design	4000	In Progress	5/3/2026	15/5/2026	Nawabshah	LinkedIn	Paid	4.8

Duplicate Records Check

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24	C024	Disha	SmartEdge Technologies	Graphic Design	9000	Pending	3/3/2026	10/5/2026	Sukkur	Website	Pending	4.2

Pivot Table Analysis

Pivot tables were created to summarize business data efficiently.

The following pivot tables were prepared:

- Total Budget by Service
- Number of Clients by City
- Project Status Count

- Lead Source Performance
- Payment Status Summary
- Average Rating by Service

These pivot tables helped identify business trends and performance across different categories.

Pivot Tables

project status count	
Row Label	Count of Project_Status
Completed	16
In Progress	20
Pending	14
Grand Total	50

lead source performance	
Row Label	Count of Lead_Source
Facebook	10
Instagram	9
LinkedIn	12
Referral	10
Website	9
Grand Total	50

total budget by service	
Row Labels	Sum of Project_Budget
Data Analytics	\$62,500.00
Graphic Design	\$61,600.00
Mobile App Development	\$42,000.00
SEO Services	\$34,000.00
UI/UX Design	\$45,100.00
Web Development	\$52,700.00
Grand Total	\$297,900.00

payment status summary	
Row Label	Count of Payment_Status
Paid	24
Pending	26
Grand Total	50

average rating by service	
Row Labels	Sum of Client_Rating
Data Analytics	41
Graphic Design	48.5
Mobile App Development	26.3
SEO Services	21.8
UI/UX Design	39.7
Web Development	37.1
Grand Total	214.4

Charts and Data Visualization

Various charts were developed to visualize business performance.

Charts included:

- Revenue by Service (Bar Chart)
- Project Status Distribution (Pie Chart)
- Clients by City (Column Chart)
- Projects by Date (Line Chart)
- Payment Status (Donut Chart)
- Budget and Rating by Service (Combo Chart)

These visualizations make the business data easier to understand and compare.

Charts



Power BI Dashboard Explanation

A Power BI dashboard was developed to provide an interactive overview of business performance.

The dashboard includes:

- Revenue Analysis
- Client Analysis
- Service Performance
- Payment Status
- Project Status
- Interactive Filters

Users can filter the dashboard using different categories to quickly analyze business performance.

Power BI Dashboard



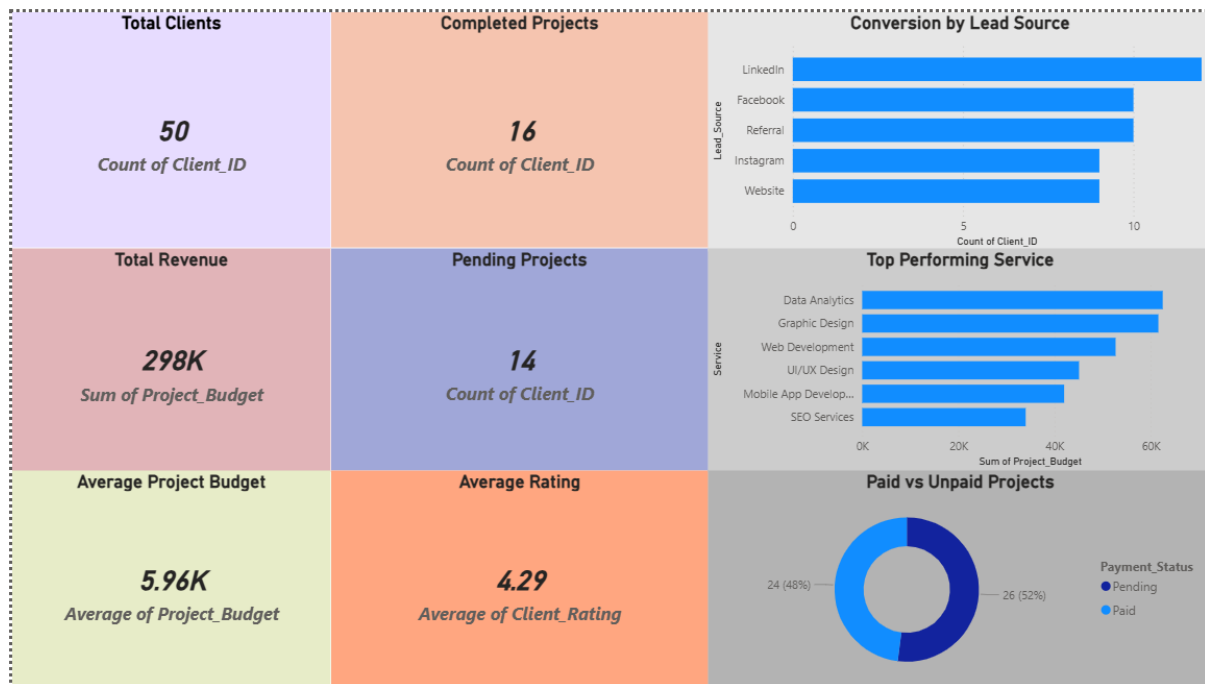
KPI Dashboard Explanation

A KPI dashboard was created to monitor the most important business metrics.

KPIs Included:

- Total Clients
- Total Revenue
- Average Project Budget
- Completed Projects
- Pending Projects
- Average Rating
- Top Performing Service
- Conversion by Lead Source
- Paid vs Unpaid Projects

KPI Dashboard



Business Insights

1. CoreTech Innovation has successfully managed 50 client projects.
2. Total business revenue is approximately 298K.
3. Data Analytics is the highest revenue-generating service.
4. LinkedIn is the best-performing lead source.
5. The average project budget is approximately 5.96K.
6. The average client rating is 4.29, indicating high customer satisfaction.
7. Some projects are still pending and require closer monitoring.
8. Unpaid projects indicate a need for better payment follow-up.

Recommendations

1. Invest more resources in Data Analytics services.
2. Increase marketing efforts on LinkedIn.
3. Improve project tracking to reduce pending projects.
4. Strengthen payment collection procedures.

5. Promote lower-performing services through targeted marketing campaigns.

Conclusion

This project successfully analyzed CoreTech Innovation's business performance using Excel, MySQL, and Power BI. The dashboards and visualizations provide valuable insights into revenue, clients, services, project status, and payment performance. The results can help management make informed business decisions and improve future business growth.

Demo Video Link

https://drive.google.com/file/d/11_oeLQJrwbTCHKvxrLiHUxrlzrMH_qBo/view?usp=drive_link